THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS FULL SYLLABUS PAPER SET 01

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The 4th triangular number is: 1

(a) 11 **(b)** 8 **(c)** 14 **(d)** 10

2. Hexagonal numbers 1, 7, 19, 37, … are formed by: 1

(a) dots in a triangle
(b) dots in a square grid
(c) dots arranged in hexagons around a centre
(d) dots in a line

3. Two lines that cross at a point are: 1

(a) intersecting lines **(b)** segments
(c) rays **(d)** parallel lines

4. An angle of 300° is: 1

(a) acute **(b)** straight **(c)** obtuse **(d)** reflex

5. The smallest 3-digit palindrome is: 1

(a) 101 **(b)** 121 **(c)** 111 **(d)** 100

6. In the Collatz process, if the number is even we: 1

(a) subtract 1 **(b)** add 1 **(c)** halve it **(d)** triple it

7. A table showing how many times each item occurs is a: 1

(a) multiplication table **(b)** time table
(c) frequency table **(d)** pictograph

8. A graph that uses pictures or symbols to show data is a: 1

(a) line graph **(b)** pie chart **(c)** pictograph **(d)** bar graph

9. The HCF of 15 and 25 is: 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 10

10. The only even prime number is: 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** 4

11. Area of a rectangle of length 18 cm and breadth 11 cm is: 1

(a) 198 cm² **(b)** 29 cm² **(c)** 216 cm² **(d)** 58 cm²

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12. The area of a triangle is: 1

(a) base × height **(b)** $2\times$ base × height

(c) base + height **(d)** $\frac{1}{2}\times$ base × height

13. Half of a pizza is eaten and one fourth is eaten more. The fraction eaten is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{4}$

14. $\frac{1}{2} + \frac{1}{3}$ equals: 1

(a) $\frac{11}{12}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{47}{60}$ **(d)** $\frac{5}{6}$

15. The distance of a point on the circle from its centre is called its: 1

(a) tangent **(b)** diameter **(c)** chord **(d)** radius

16. A square of side 6 cm has diagonals each: 1

(a) shorter than 6 cm **(b)** equal to 3 cm

(c) longer than 6 cm **(d)** equal to 6 cm

17. Symmetry is commonly seen in: 1

(a) rangoli and architecture **(b)** only fractions

(c) only prime numbers **(d)** only bar graphs

18. The angle of rotational symmetry of a regular octagon is: 1

(a) 45° **(b)** 60° **(c)** 90° **(d)** 30°

19. If the temperature falls from 2°C by 6°C it becomes: 1

(a) −4°C **(b)** 8°C **(c)** −8°C **(d)** 4°C

20. A deposit of ₹100 and a withdrawal of ₹100 together give: 1

(a) ₹100 **(b)** −₹200 **(c)** 0 **(d)** ₹200

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. What do we get on adding 1+2+1, 1+2+3+2+1, … ? 2

22. Write the next two numbers: 3, 6, 12, 24, … 2

23. How many symbols are needed for 85 if one symbol stands for 5? 2

24. Is 128 divisible by 4? Show by the rule. 2

25. Which needs the perimeter: fencing a field or tiling a floor? 2

26. Arrange in ascending order: $\frac{1}{3}, \frac{1}{2}, \frac{1}{6}$. 2

27. How many right angles does a square have? 2

28. Find the distance between −7 and 3 on the number line. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Find the 15th odd number and the sum of the first 15 odd numbers. 3

30. Write the number of degrees made by a turn of 1/4, 1/2 and 3/4 of a circle. 3

31. Draw a pictograph for: Mon 15, Tue 25, Wed 20 using 1 symbol = 5. 3

32. A rectangle has area 36 cm². List three possible pairs of whole-number sides and the perimeters. 3

33. Draw an artwork using circles with the same centre and different radii. Name the pattern. 3

34. Complete the figure: if one half of a symmetric figure is a right triangle with the mirror along the leg, what is the full figure? 3

35. Find the HCF of 36, 48 and 60. 3

OR

A number is divisible by both 4 and 6. Is it divisible by 24? Explain with an example. 3

36. Explain with an example why $(-a) + a = 0$. 3

OR

Show on a number line: $-2 + 5$. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) Describe an experiment using paper folding to get a right angle. 5
(ii) Use it to find an angle of 45°.

OR

(i) Find the angle through which the hour hand of a clock turns from 3 p.m. to 7 p.m. 5
(ii) the minute hand from 3:00 to 3:40.

38. (i) What is a 4-digit palindrome pattern? 5
(ii) Show that every 4-digit palindrome is divisible by 11.
(iii) Test with 3443.

OR

(i) Write the rule for digit sum. 5
(ii) Find the digit sum of 12, 345.
(iii) Test divisibility by 3 for 12345.

39. (i) Write $\frac{3}{4}$ of a day in hours. 5
(ii) Write $\frac{2}{3}$ of an hour in minutes.
(iii) What is $\frac{5}{8}$ of 1 kg in grams?

OR

(i) Compare $\frac{7}{3}$ and $2\frac{1}{2}$. 5
(ii) Add them.
(iii) Write the sum as a mixed fraction.

40. (i) Explain why a circle has infinite lines of symmetry. 5
(ii) Find its order of rotation.
(iii) How does it differ from polygons?

OR

(i) Show that the mirror image of the letter ‘E’ in a horizontal mirror is the same. 5
(ii) Give two other letters.

END OF QUESTION PAPER — Total: 80 Marks

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